

Chapter 1. Basic

No. _____

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1. 事件间的关系: ① 互不相容: $A \cap B = \emptyset$ (互斥)

② 对立: $A \cap B = \emptyset$ 且 $A \cup B = \Omega$.
(互为逆事件)

③ 独立:

当 $P(B) > 0$, $P(A|B) = P(A)$

或 $P(AB) = P(A) \cdot P(B)$

若 A, B 独立

则

$P(A), P(B) > 0$ 则

AB 一定相容.

2. 概率性质: ① (加法): $P(A \cup B) = P(A) + P(B) - P(AB)$

$$\Rightarrow P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(BC) - P(AC) + P(ABC)$$

3. 随机试验模型: ① 古典概型: $P(A) = \frac{k}{n}$

② 几何概型: $P(A) = \frac{G \text{ 的测度}}{\Omega \text{ 的测度}}$ 测度 (长度, 面积, 体积等)

4. 全概率公式: $P(A) = \sum P(AB_k)$

$$= \sum P(A|B_k) \cdot P(B_k)$$

5. 贝叶斯公式: $P(B_i|A) = \frac{P(A|B_i) \cdot P(B_i)}{\sum P(A|B_k) \cdot P(B_k)}$

$\rightarrow$ 全概率

e.g. P11. 投针与全圆问题.

No.

(10')

Chapter 2. Discret r.v.

DATE / /

1. 一维离散 r.v. 分布律: 非负性 + 规范性.

2. 常见分布: ① 0-1 分布:

$$P\{X=k\} = p^k (1-p)^{1-k}$$

② 二项分布: (n 重贝努利试验). 两态独立重复, 其中一态出现数.

$$P\{X=k\} = C_n^k p^k (1-p)^{n-k}$$

$$X \sim B(n, p)$$

③ 泊松分布: (可加性)

$$P\{X=k\} = \frac{\lambda^k}{k!} e^{-\lambda}$$

$$X \sim P(\lambda)$$

④ 几何分布: Δ 无记忆性: $P(X > n+m | X > n) = P(X > m)$. P_{40} 解长问题.

$$P\{X=k\} = (1-p)^{k-1}$$

$$X \sim G(p)$$

3. 边缘分布律:

$X \backslash Y$	Y_1	Y_2	$P_{i\cdot}$
X_1	p_{11}	p_{12}	$p_{1\cdot}$
X_2	p_{21}	p_{22}	$p_{2\cdot}$
$P_{\cdot j}$	$p_{\cdot 1}$	$p_{\cdot 2}$	1

Chapter 3. Continuous r.v.

No. (301)

DATE / /

1. 分布函数 $F(x)$: ① 单调不减性
② 非负规范性.
③ 右连续性. $\rightarrow 0^+$

$$F(x) = P\{X \leq x\} = \int_{-\infty}^x \underbrace{f(t)}_{\text{密度函数}} dt.$$

2. 常见分布: ① 均匀分布 $X \sim U(a, b)$

$$f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{else} \end{cases}$$

- ② 指数分布 $X \sim E(\lambda)$

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

- ③ 正态分布: $X \sim N(\mu, \sigma^2)$ (可加性)

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Δ 分布函数: $F(x) = P\{X \leq x\} \rightarrow$ 标准正态.

Δ α 分位点: $P\{X > z_\alpha\} = \alpha \Rightarrow z_\alpha$ 为标准正态分布的上侧 α 分位点.

3. 2-D 连续 r.v. 分布函数:

$$F(x, y) = \int_{-\infty}^x \int_{-\infty}^y \underbrace{f(u, v)}_{\text{联合密度函数}} du dv.$$

边缘密度函数:

$$F_X(x) = \int_{-\infty}^x \left(\int_{-\infty}^{+\infty} f(u, y) dy \right) du.$$

条件密度函数:

$$F_{X|Y}(x|y) = \int_{-\infty}^x \frac{f(u, y)}{f_Y(y)} du.$$

4. X, Y 独立 $\Leftrightarrow f(x, y) = f_X(x) \cdot f_Y(y)$

5. 求 $Y = g(X)$ 的密度函数: ① 分布函数法: $F_Y(y) = P\{Y \leq y\} = P\{g(X) \leq y\} = \int_{g(x) \leq y} f(x) dx$

$$\Rightarrow f_Y(y) = \frac{dF_Y(y)}{dy}.$$

② PPT. Chap 3. Prob.

② 公式法 1: $y = g(x)$ 在 $(-\infty, +\infty)$ 上严格单调. 可导. $\Rightarrow f_Y(y) = \begin{cases} |f[h(y)] \cdot |h'(y)|| & a < y < b \\ 0 & \text{else.} \end{cases}$

$h(y)$ 为 $g(x)$ 的反函数.

③ 公式法 2: $f_X(x) = f_X * f_Y = \int_{-\infty}^{+\infty} f_X(x) \cdot f_Y(z-x) dx = \int_{-\infty}^{+\infty} f_X(z-y) f_Y(y) dy.$
(卷积)

Chapter 4. Numerical characteristics of r.v.

No. (10+5)

DATE 10/10/2020

1. 数学期望: $E(X) = \int_{-\infty}^{+\infty} x f(x) dx$

如果 $\int_{-\infty}^{+\infty} |x| f(x) dx < +\infty$, 则

△ 分布: 离散: ① 0-1 分布. $E(X) = P$. $D(X) = P(1-P)$

② 二项分布. np $np(1-p)$

③ 泊松... λ λ

④ 几何... $\frac{1}{p}$

连续: ① 均匀... $\frac{a+b}{2}$ $\frac{(b-a)^2}{12}$

② 指数... $\frac{1}{\lambda}$ $\frac{1}{\lambda^2}$

③ 正态... μ σ^2

2. $\Gamma(\alpha) = \int_0^{+\infty} t^{\alpha-1} \cdot e^{-t} dt$: ① $\Gamma(\alpha+1) = \alpha \Gamma(\alpha)$

② $\Gamma(n+1) = n!$. $n \in \mathbb{N}^+$

③ $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

3. if $\int_{-\infty}^{+\infty} |g(x)| \cdot f(x) dx < \infty$, 则 $E(Y) = E[h(X)] = \int_{-\infty}^{+\infty} h(x) f(x) dx$

$Y = h(X)$ 复合函数

$\Rightarrow E(Z) = E[h(X, Y)] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h(x, y) \cdot f(x, y) dx dy$

(离散: $E(Z) = \sum_{i,j} h(x_i, y_j) \cdot p_{ij}$)

4. 方差: $D(X) = E[X - E(X)]^2 = E(X^2) - E^2(X)$

① $D(cX) = c^2 D(X)$

② $D(X \pm Y) = D(X) + D(Y) \pm 2 \text{Cov}(X, Y)$

5. 协方差: $\text{Cov}(X, Y) = E\{[X - E(X)] \cdot [Y - E(Y)]\} = E(XY) - E(X) \cdot E(Y)$

① $\text{Cov}(aX+b, cY+d)$

② $\text{Cov}(X_1+X_2, Y)$

(性质?) $= \text{Cov}(X_1, Y) + \text{Cov}(X_2, Y)$

6. 相关系数: $\rho = \frac{\text{Cov}(X, Y)}{\sqrt{D(X)} \cdot \sqrt{D(Y)}}$ ($|\rho| = 1 \Rightarrow$ 线性相关)

③ $\text{Cov}(X, X) = D(X)$

$\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$

Chapter 6. Mathematical statistics basic concepts.

1. 统计量: ① 样本均值: $\bar{X} = \frac{1}{n} \sum X_i$

② 样本方差: $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2 = \frac{1}{n-1} (\sum X_i^2 - n\bar{X}^2)$

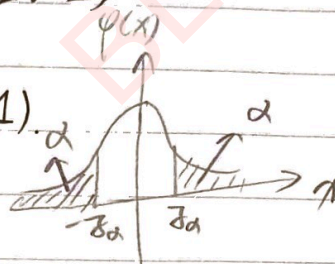
③ k 阶原点矩: $A_k = \frac{1}{n} \sum X_i^k$

④ k 阶中心矩: $B_k = \frac{1}{n} \sum (X_i - \bar{X})^k$

2. 抽样分布: ① 标准正态: $N(0, 1)$

$$P(Z > z_\alpha) = \alpha$$

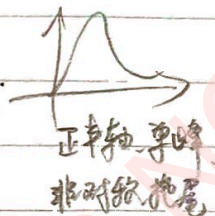
上侧 α 分位点



② χ^2 分布: $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(0, 1)$

$$\Rightarrow X_1^2 + X_2^2 + \dots + X_n^2 = \sum_{i=1}^n X_i^2 \sim \chi^2(n)$$

$$(E(\chi^2) = n, D(\chi^2) = 2n)$$



③ t 分布: $X \sim N(0, 1), Y \sim \chi^2(n), X$ 与 Y 独立.

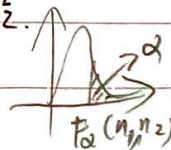
$$\Rightarrow t = \frac{X}{\sqrt{Y/n}} \sim t(n)$$

$$(E(t) = 0, D(t) = \frac{n}{n-2})$$



④ F 分布: $X \sim \chi^2(n_1), Y \sim \chi^2(n_2), X$ 与 Y 独立.

$$\Rightarrow F = \frac{X/n_1}{Y/n_2} \sim F(n_1, n_2)$$



$$F: F_{1-\alpha}(n_1, n_2) = \frac{1}{F_\alpha(n_1, n_2)}$$

3. 定理: $X_1, X_2, \dots, X_n$ 是总体 $N(\mu, \sigma^2)$ 的容量为 n 的样本.

$$\bar{X} = \frac{1}{n} \sum X_i, S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$$

$$(1). \bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \Rightarrow \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

$$(2). \frac{n-1}{\sigma^2} S^2 = \frac{1}{\sigma^2} \sum (X_i - \bar{X})^2 \sim \chi^2(n-1)$$

(3). $\bar{X}$ 与 S^2 相互独立.

$$(4). \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

Chapter 7. Parameter Estimation. (201)

DATE / /

e.g. 一天电话数 $\sim P(\lambda)$.

1. 点估计. ① 矩估计法: 步骤: 1. 计算总体矩: $E(X) = \lambda$.

e.g. $X \sim N(\mu, \sigma^2)$ $X_1, X_2, \dots, X_n$ 为样本 2. 矩估计: $\hat{\mu} = A_1$

简单随机样本 求 $\hat{\mu}$ 和 $\hat{\sigma}^2$.

3. 将矩表达式代入估计式: $\lambda = A_1$

1. $\mu_1 = E(X) = \mu$, $\mu_2 = E(X^2)$

4. 总结: $\hat{\lambda}_m = A_1$

$$= D(X) + E(X)^2$$

代值 $\Rightarrow \hat{\lambda} = 1.22$

$$= \sigma^2 + \mu^2$$

2. 令 $\mu_1 = A_1$

$\mu_2 = A_2$

3. $\mu = A_1$

$$\sigma^2 + \mu^2 = A_2$$

$$4. \begin{cases} \hat{\mu}_m = A_1 \\ \hat{\sigma}_m^2 = A_2 - A_1^2 = \frac{1}{n} \sum (X_i - \bar{X})^2 = B_2 \end{cases}$$

② 最大似然估计法: 步骤: 1. 设 $x_1, x_2, \dots, x_n$ 为一组样本观测值.
(likelihood)

2. 构造似然函数:

e.g. 同上.

1. 设 -- 样本观测值.

$$\text{已知 } P(X=x) = \frac{\lambda^x}{x!} \cdot e^{-\lambda}$$

$$L(\lambda) = \prod_{i=1}^n P(X=x_i) = \prod_{i=1}^n \frac{\lambda^{x_i}}{x_i!} \cdot e^{-\lambda} = \left(\prod_{i=1}^n \frac{1}{x_i!} \right) \lambda^{\sum x_i} e^{-n\lambda}$$

$$2. L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x_i - \mu)^2}{2\sigma^2}} = \frac{1}{(\sqrt{2\pi}\sigma)^n} \cdot e^{-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2}$$

3. 构造似然方程:

$$3. \frac{\partial L(\mu, \sigma^2)}{\partial \mu} = \frac{1}{\sigma^2} \sum (x_i - \mu)$$

$$\frac{d \ln L(\lambda)}{d \lambda} = \sum_{i=1}^n \left(\frac{x_i}{\lambda} - 1 \right) = 0$$

$$\frac{\partial L(\mu, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (x_i - \mu)^2 = 0$$

4. 总结: 极大似然估计值: $\hat{\lambda}_m = \bar{x}$

极大似然估计量: $\hat{\lambda}_L = \bar{X}$

$$4. \hat{\mu}_L = \bar{x}, \hat{\sigma}_L^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

$$\Rightarrow \hat{\mu}_L = \bar{X}, \hat{\sigma}_L^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$$

2. 评判标准. ① 无偏性: $E(\hat{\theta}) = \theta$.

e.g. $X_1, X_2, \dots, X_n$ 来自 $N(\mu, \sigma^2)$

e.g. $X \sim U(0, \theta)$.

$$1) E(\bar{X}) = E\left(\frac{1}{n} \sum X_i\right) = \frac{1}{n} \sum \mu = \mu \quad \text{无偏}$$

$$\hat{\theta}_L = \sum_{i=1}^n x_i$$

$$f_{(n)}(x) = n F(x)^{n-1} f(x) = \begin{cases} \frac{n}{\theta^n} x^{n-1} & 0 < x < \theta \\ 0 & \text{else} \end{cases}$$

$$2) \frac{n-1}{\sigma^2} S^2 \sim \chi^2(n-1), E\left(\frac{n-1}{\sigma^2} S^2\right) = n-1$$

$$\Rightarrow E(S^2) = \sigma^2, E(B_2) = E\left(\frac{n-1}{n} S^2\right) = \frac{n-1}{n} \sigma^2 \quad \text{有偏}$$

$$E(\sum_{i=1}^n x_i) = \int_0^\theta x f_{(n)}(x) dx = \frac{n}{n+1} \theta \quad \text{有偏}$$

或者期望性质法.

② 有偏性: $D(\bar{X})$ 越小越好.

3. 区间估计: 区间 $(\underline{\theta}, \bar{\theta})$ 为参数 θ 的置信水平为 $1-\alpha$ 的置信区间.

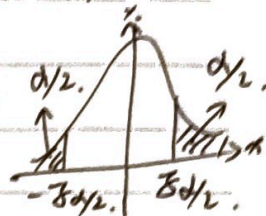
枢轴量法.

e.g. 设总体 $X \sim N(\mu, \sigma^2)$, $X_1, X_2, \dots, X_n$ 为来自总体的样本, 对样本值为 $x_1, x_2, \dots, x_n$.

1. 求 μ 的置信度 $1-\alpha$ 的置信区间.

① σ^2 已知: $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \Rightarrow Z \in (-z_{\alpha/2}, z_{\alpha/2})$

$\Rightarrow (\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}})$



② σ^2 未知: $T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$

$\Rightarrow (\bar{X} - t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}})$

2. 求 σ^2 .

① μ 已知: $\chi^2 = \frac{1}{\sigma^2} \sum (X_i - \mu)^2 \sim \chi^2(n)$

$\Rightarrow \left(\sqrt{\frac{\sum (X_i - \mu)^2}{\chi_{\alpha/2}^2(n)}}, \sqrt{\frac{\sum (X_i - \mu)^2}{\chi_{1-\alpha/2}^2(n)}} \right)$

② μ 未知: $\chi^2 = \frac{n-1}{\sigma^2} S^2 \sim \chi^2(n-1)$

$\Rightarrow \left(\sqrt{\frac{(n-1)S^2}{\chi_{\alpha/2}^2(n-1)}}, \sqrt{\frac{(n-1)S^2}{\chi_{1-\alpha/2}^2(n-1)}} \right)$

Chapter 8. Hypothesis Testing

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1. Basic ① H_0 : 原假设 (零假设) H_1 : 备择假设.

② 拒绝域: $|\bar{x} - \mu_0| > k$ (小概率原理)

③ 在真 (零假设) $\alpha = P\{\text{拒绝 } H_0 | H_0 \text{ 为真}\}$

取伪: $\beta = P\{\text{接受 } H_0 | H_0 \text{ 为假}\}$

$\alpha \downarrow$ 则 $\beta \uparrow$

default: $\alpha = 0.05$ (显著性水平) $\rightarrow$ 显著性检验

提出 H_0, H_1

α, k

样本判断

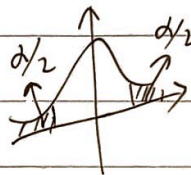
(单正态总体均值的 HT.)

e.g. $\alpha = P\{|\bar{x} - \mu_0| > k | \mu = \mu_0\}$

当 $\mu = \mu_0$ 时 $d = P\{|\bar{x} - \mu_0| > k\}$

$\bar{x} \sim N(\mu_0, \sigma^2/n)$

标准化: $\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$



$$\alpha = P\left\{\left|\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}\right| > \frac{k}{\sigma/\sqrt{n}} \mid \mu = \mu_0\right\}$$

$$\Rightarrow \frac{k}{\sigma/\sqrt{n}} = z_{\alpha/2}$$

Δ 故拒绝域: $\left|\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}\right| > z_{\alpha/2}$

(单正态总体方差的 HT.)

例

e.g. $H_0: \mu = \mu_0$ $H_1: \mu \neq \mu_0$

由于 σ^2 已知, 构造统计量:

$$U = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \stackrel{H_0 \text{ 为真}}{\sim} N(0, 1)$$

由 $\alpha = P\{\text{拒绝 } H_0 | H_0 \text{ 为真}\}$ 得

拒绝域: $|U| > z_{\alpha/2}$

其中 $\bar{x} = 31.13$, $\mu_0 = 32.5$, $\sigma = 1.1$, $n = 6$.

$$\text{计算得 } |U| = \left| \frac{31.13 - 32.5}{1.1/\sqrt{6}} \right| = 3.05$$

$$\text{查表得 } z_{\alpha/2} = z_{0.025} = 1.96$$

由于 $|U| > z_{\alpha/2}$ 故拒绝 H_0 .

认为平均强度不为 32.5.